Lab - Investigate BIOS or UEFI Settings

Introduction

In this lab, you will boot the computer, explore the firmware setup utility program, and change the boot order sequence.

Recommended Equipment

- Computer with or without operating system
- Motherboard manual

Instructions

Part 1: Enter BIOS or UEFI.

Step 1: Power on the computer.

a. Plug the power supply cable into an AC wall outlet.

b. If there is a power switch on the power supply, set the switch to “1” or “on”.

c. Turn on the computer with the power button on the front panel.

Note: If the computer beeps more than once, or if the power does not come on, notify your instructor.

Step 2: Enter the firmware setup program.

During POST, press the firmware setup key or key combination. The firmware setup utility program screen will appear.

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Questions:

What is the key or combination of keys used to enter the firmware setup utility program?

The firmware setup utility program is commonly accessed by pressing the Delete (Del) or F2 key during the computer's startup process. The exact key depends on the motherboard or computer manufacturer, but these are the most used keys.

Who manufactures the BIOS or UEFI system for your computer?

The BIOS or UEFI firmware is commonly manufactured by companies such as American Megatrends (AMI), Phoenix Technologies, or Insyde Software. The manufacturer is displayed on the BIOS or UEFI main screen during startup.

.What is the BIOS or UEFI version?

The BIOS or UEFI version is displayed on the main page of the firmware setup utility. An example is Version 2.20, but the version number will vary depending on the computer or motherboard being used

Part 2: Explore the Settings.

Step 1: List the main menu options.

Question:

List the main menu options and describe what is monitored in each menu?

The BIOS or UEFI setup utility typically includes the Main, Advanced, Security, Boot, Power, and Save & Exit menus. The Main menu displays system information such as the date, time, processor, and memory. The Advanced menu contains hardware configuration settings. The Security menu manages passwords and security features. The Boot menu controls the boot device order. The Power menu contains power management options, while the Save & Exit menu allows users to save changes or restore default settings.

Step 2: Find the security settings.

Navigate through each screen to find the security settings.

Question:

What security settings and features are available?

The Security menu usually includes options such as setting an administrator password, user password, Secure Boot, Trusted Platform Module (TPM), hard drive passwords, and BIOS write protection. These features help protect the computer from unauthorized access and improve system security.

Step 3: Find the CPU settings.

Navigate through each screen to find the CPU settings.

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Questions:

What is the CPU speed?

The CPU speed is displayed in the BIOS or UEFI information page. A typical processor may have a speed of 3.40 GHz, although the exact value depends on the installed processor.

What other information is listed for the CPU?

The BIOS or UEFI also displays additional processor information, including the processor model, number of cores, number of threads, cache size, virtualization support, and sometimes the current CPU temperature and operating frequency.

Step 4: Find the RAM settings.

Navigate through each screen to find the RAM settings.

Questions:

What is the RAM speed?

The RAM speed is shown in the memory information section of the BIOS or UEFI. A common memory speed is 3200 MHz, but the exact speed depends on the installed RAM modules.

What other information is listed for the RAM?

The BIOS or UEFI displays information such as the total installed memory, memory type (DDR4 or DDR5), number of memory slots used, memory capacity per slot, and the current operating frequency of the RAM.

Step 5: Find the hard drive settings.

Navigate through each screen to find the hard drive settings.

Questions:

What information is listed for the hard drive?

The hard drive information includes the storage device model, total storage capacity, interface type such as SATA or NVMe, serial number, SMART status, and whether the drive is recognized as a bootable device.

Step 6: Find the boot order sequence.

Navigate through each screen to find the boot order sequence.

Questions:

What is the first boot device in the boot order sequence?

For this activity, the first boot device should be set to the optical drive (CD/DVD drive) so that the computer can boot from installation or recovery media if needed.

How many additional devices can be assigned in the boot order sequence?

Most BIOS or UEFI systems allow several additional devices to be assigned in the boot order. Typically, there are three to five more boot devices, such as the hard drive, USB drive, network boot, or other storage device

Step 7: Set the device boot order settings.

a. Ensure that the first boot order device is the optical drive.

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b. Ensure that the second boot order device is the hard disk drive.

Questions:

Why would you change the first boot device to the optical drive?

The first boot device is changed to the optical drive so the computer can boot from a bootable DVD or CD. This is useful when installing a new operating system, running diagnostic tools, or repairing an existing operating system.

What happens when the computer boots and the optical drive does not contain bootable media?

If the optical drive does not contain a bootable disc, the BIOS or UEFI automatically checks the next device in the boot order, such as the hard drive or SSD, and attempts to boot the operating system from that device.

Step 8: Find the power management setup or ACPI screen.

Navigate through each screen to find the power management setup screen, or ACPI screen.

Question:

What power management settings are available?

The Power or ACPI menu typically includes settings for sleep modes, wake-on-LAN, automatic power recovery after a power failure, CPU power management, and other energy-saving features that help reduce power consumption.

Step 9: Find the PnP settings.

Navigate through each screen to find the PnP settings.

Question:

What PnP settings are available?

The Plug and Play (PnP) settings usually include options for Plug and Play operating system support, PCI device configuration, PCI Express settings, resource allocation, and legacy USB support to ensure hardware devices are detected and configured correctly.

Step 10: Find the splash screen settings.

Navigate through each screen to find the splash screen settings.

Question:

What splash screen settings are available?

The splash screen settings typically allow users to enable or disable the full-screen manufacturer logo during startup, enable Quiet Boot, or display detailed POST messages instead of the manufacturer's logo. These settings let users choose how startup information is displayed before the operating system loads

Step 11: Save and exit the setup utility program.

Save the new BIOS/UEFI settings and exit the setup utility program. The computer should restart automatically.

Note: An error message stating that an OS cannot be found (or a similar error) will appear on the screen after the computer boots. An operating system must now be installed to prevent this error. It is safe to turn off the computer at this time.

This lab is complete. Please have the instructor verify your work.

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